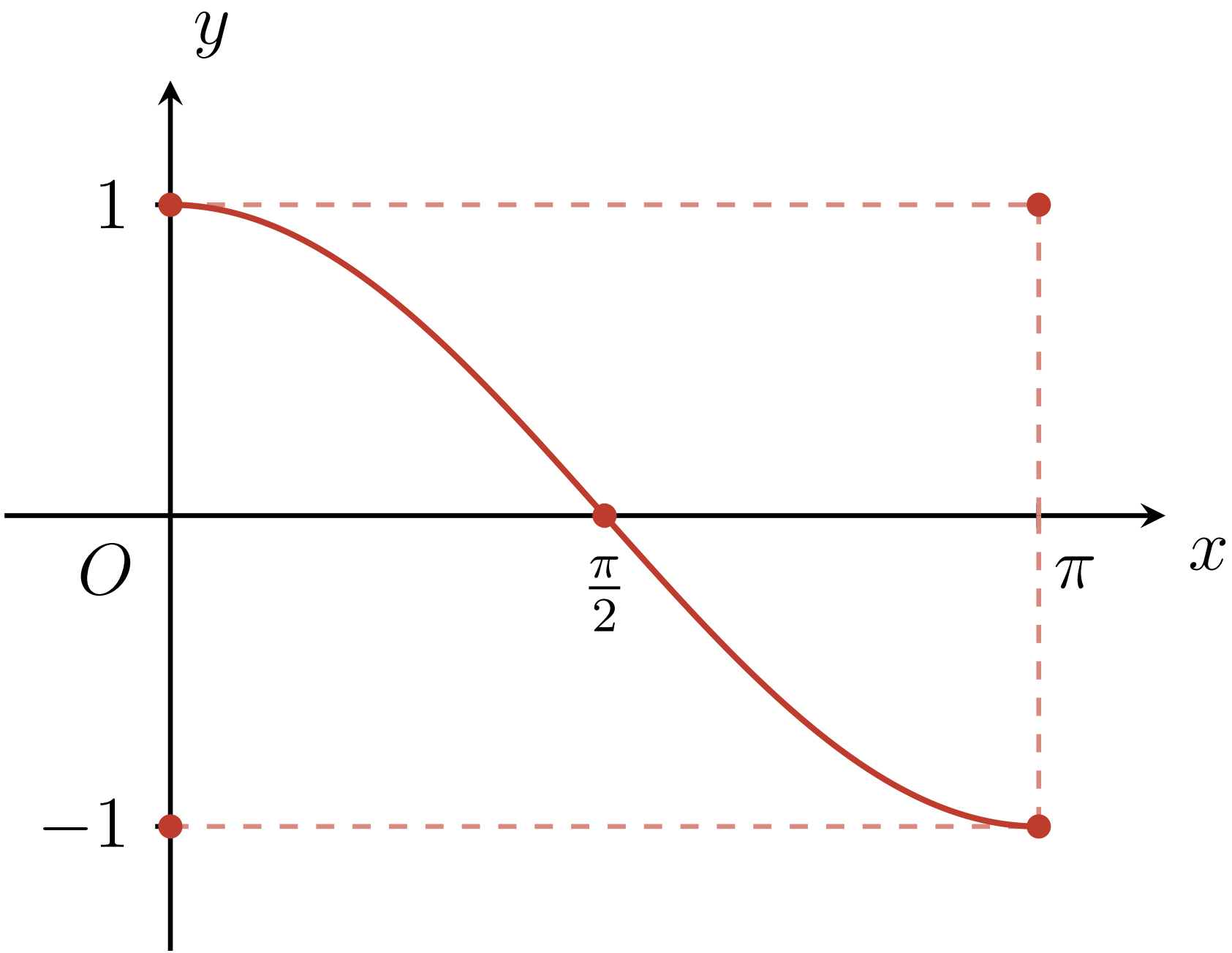

$$y = \cos(x), \quad 0 \leq x \leq \pi$$